Genetic information and molecular pathways recapitulate psychiatric drug prescriptions and diagnoses

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Introduction

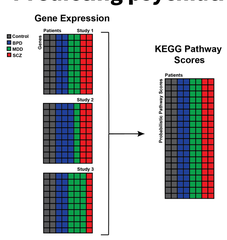
- **1 in 5** experience mental illness annually
- Current treatments are often **ineffective**
- Psych disorders have **complex genetics**
- Diagnosis from **overlapping symptoms**
- **Disruption in transcription** implicated

Goals

- **Diagnose** psychiatric disease from gene expression
- **Rank** psychiatric drugs by disease from drug gene targets and disease gene sets
- **Evaluate** rankings with clinical prescribing patterns from the UK Biobank

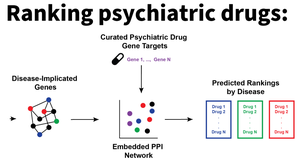
Methods

Predicting psychiatric disease:



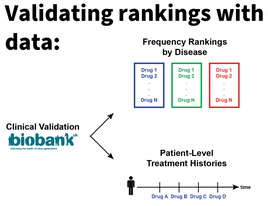
- Obtain **gene expression** of patients with Bipolar, Depression, and Schizophrenia
- Use **probabilistic pathway score (PROPS)** to generate KEGG pathway scores from gene expression
- Classify patients from pathways scores with **Random Forest** model

Ranking psychiatric drugs:



- Curate **disease genes** and psychiatric **drug targets**
- Embed STRING v10 PPI Network with **node2vec**
- **Rank drugs** by highest "overlap" of drug sets

Validating rankings with clinical data:



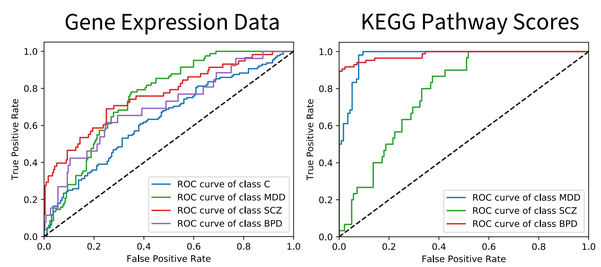
- Create **drug frequency ranks**
- Compare to predicted rankings
- Extract patient **trajectories**
- Assess change in rank over time

Predicting Psychiatric Disease

- **KEGG pathway scores as features** improves accuracy compared to batch-corrected gene expression from 56% to 78%
- AUROC for Multiple Classifiers

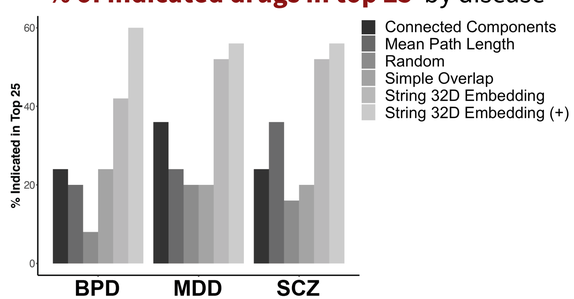
Classifier	AUROC	
	Gene Expression	Pathway Scores
Random Forest	0.66 +/- 0.01	0.87 +/- 0.02
Decision Tree	0.57 +/- 0.02	0.80 +/- 0.01
Support Vector	0.52 +/- 0.03	0.83 +/- 0.02

- ROC for One-vs-Rest Random Forest Model



Ranking Psychiatric Drugs

- Curated disease-associated genes and gene targets for **167 psychiatric drugs**
- Assessed "**overlap**" between disease and disease gene sets
- **% of indicated drugs in top 25** by disease



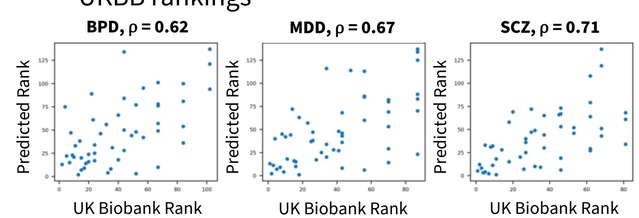
- **Embedding method** (mean cosine similarity) performed better by capturing latent neighborhood structural features of the networks

Validating Rankings with Clinical Data

- Extracted medication lists and primary and secondary ICD diagnoses from **UK Biobank**
- Patients on psychiatric drugs and with psychiatric diagnosis:

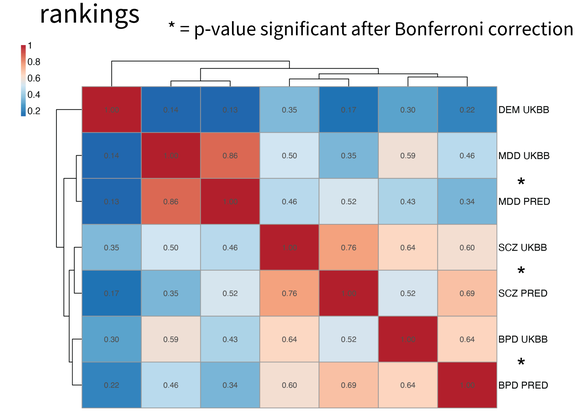
Disease	# of Pts
Bipolar Disorder	1447
Major Depressive Disorder	997
Schizophrenia	881

- Count **co-occurrences** of psychiatric diagnoses and psychiatric drug prescriptions
- Compared UKBB frequency ranks to predicted rankings with **Spearman correlation** and **Kendall's Tau**, a non-parametric hypothesis test of rank correlation
- Spearman correlation of predictions and UKBB rankings



- Kendall's Tau Matrix for predicted and UKBB rankings

* = p-value significant after Bonferroni correction



Validating Rankings with Clinical Data

- Top 5 Drugs for **Bipolar Disorder**

Predictions	UK Biobank
Maprotiline (SNRI)	Lithium (Mood stabilizer)
Carbamazepine (Na channel blocker)	Olanzapine (Atypical antipsychotic)
Lamotrigine (Na channel blocker)	Venlafaxine (SNRI)
Olanzapine (Atypical antipsychotic)	Quetiapine (Atypical antipsychotic)
Benperidol (Typical antipsychotic)	Valproate (Na channel blocker)

- Top 5 Drugs for **Major Depressive Disorder**

Predictions	UK Biobank
Buspirone (Serotonin receptor agonist)	Venlafaxine (SNRI)
Olanzapine (Atypical antipsychotic)	Mirtazapine (TCA)
Dothiepin (TCA)	Fluoxetine (SSRI)
Lorazepam (GABA receptor agonist)	Sertraline (SSRI)
Bifemelane (MOAI)	Lithium (Mood stabilizer)

- Top 5 for **Schizophrenia**

Predictions	UK Biobank
Aripiprazole (Atypical anti-psychotic)	Olanzapine (Atypical antipsychotic)
Penfluridol (First-generation anti-psychotic)	Procyclidine (Anticholinergic agent)
Mesoridazine (Neuroleptic anti-psychotic)	Risperidone (Atypical antipsychotic)
Varenicline (Acetylcholine receptor agonist)	Amisulpride (Atypical antipsychotic)
Doxepin (TCA)	Quetiapine (Atypical antipsychotic)

Conclusions

- **Probabilistic pathway scores** derived from gene expression data distinguish between psychiatric diseases
- **Embedding-based methods on PPI networks** performed the best for disease-relevant drug prioritization
- Gene-based rankings **recaptured clinical population-level prescribing patterns**